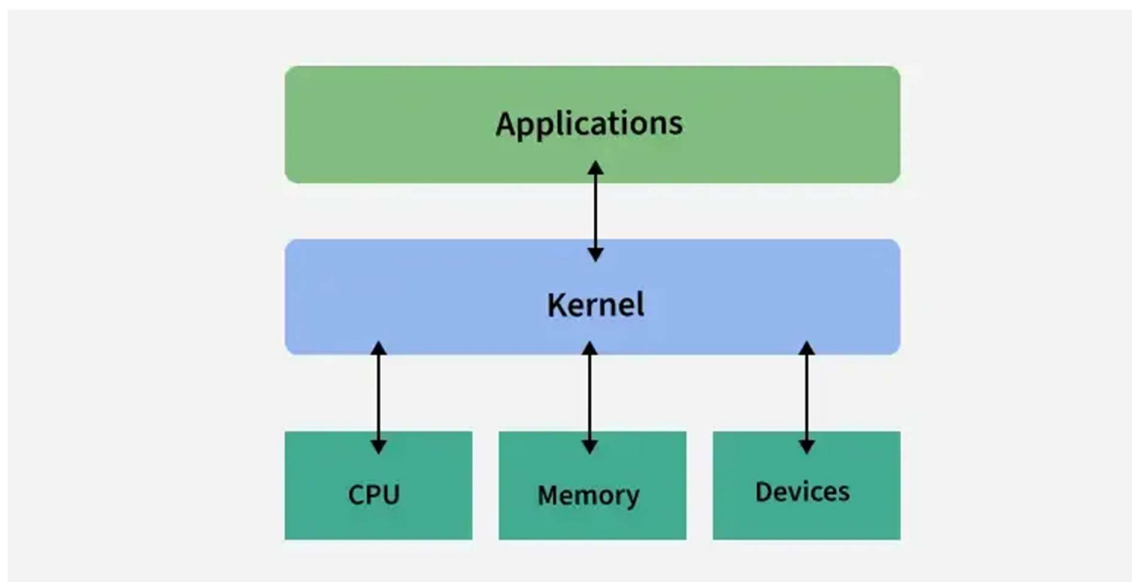


Kernel in Operating System

A kernel is the core part of an operating system. It acts as a bridge between software applications and the hardware of a computer.

- The kernel manages system resources, such as the CPU, memory and devices, ensuring everything works together smoothly and efficiently.
- It handles tasks like running programs, accessing files and connecting to devices like printers and keyboards.
- An Operating System includes the kernel as its core, but also provides a user interface, file system management, network services and various utility applications that allow users to interact with the system
- Facilitates communication between hardware and user applications.
- Ensures efficient and secure multitasking.
- Manages system stability and prevents unauthorized resource access.



Types of Kernel

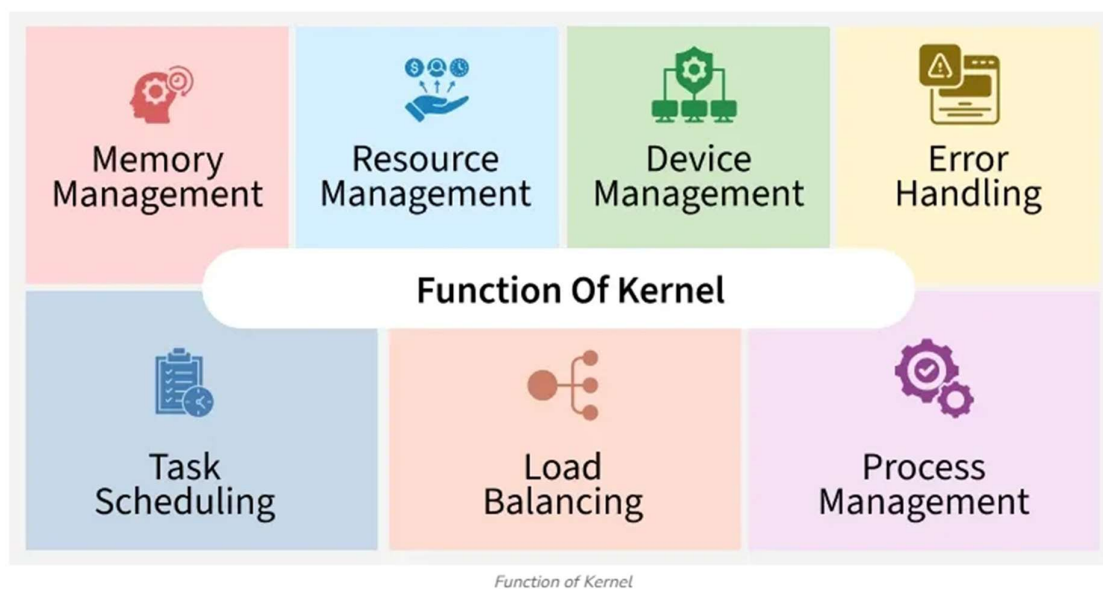
The following are different types of kernels.

- **Monolithic** kernel: all OS services run in kernel space → fast, but less fault-isolation.
Example: Unix, Linux, Open VMS, XTS-400 etc.
- **Microkernel**: minimal kernel functionality; most services moved to user space → better reliability, but more overhead.
Example: Minix 3 and Mach (true microkernel versions like Mach 3.0)

- **Hybrid kernel:** mixes monolithic + microkernel ideas; some services in kernel for speed, others isolated for safety.
Example: Windows NT family (Windows 2000, XP, Vista, 7, 8, 10 etc.), macOS / XNU, React OS and Haiku OS
- **Nanokernel:** extremely minimal kernel, providing only basic hardware abstraction; everything else outside.
Example: Nemesis and MIT Exokernel projects like XOK, Aegis
- **Exokernel:** Design where the OS handles only protection (security and safety between programs). It gives applications direct access to hardware resources instead of managing them itself.
Example: MIT Exokernel, ExOS .

Functions of Kernel

The kernel is responsible for various critical functions that ensure the smooth operation of the computer system. These functions include:



1. **Process Management** : Scheduling and execution of processes.
2. **Memory Management** : Allocation and deallocation of memory space, managing virtual memory, handling memory protection and sharing.
3. **Device Management** : Managing input/output devices, providing a unified interface for hardware devices and handling device driver communication.
4. **File System Management** : Managing file operations and providing a file system interface to applications.

5. **Resource Management** : Managing system resources (CPU time, disk space, network bandwidth). It allocating and deallocating resources as needed.
6. **Security and Access Control** : Enforcing access control policies like user permissions and authentication.
7. **Inter-Process Communication** : Facilitating communication between processes by providing mechanisms like message passing and shared memory.

Working of Kernel

- kernel is the first part of the OS loaded into memory during boot, and it stays resident while the system is running.
- It operates in a privileged mode (kernel mode), separate from user mode for applications; user apps can't directly access hardware or critical resources.
- Applications make requests to the kernel via *system calls* (or software interrupts). The kernel handles these by switching from user mode to kernel mode.
- Kernel executes the requested operation (e.g. file I/O, process creation, memory allocation).
- On completion, kernel returns result (or error) to user space.
- Kernel does context switching as needed (scheduler picks next process/thread) to allow multitasking.